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Experiment Title : Demonstration of Hub Hardware and Setting up a Basic Local Area Network (LAN)

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Experiment No. 5: Demonstration of Hub Hardware and Setting up a Basic Local Area Network (LAN)

Aim

To identify the hardware interfaces of a Network Hub and to design, configure, and verify a basic Local Area Network (LAN) using a Star Topology to observe broadcast behavior in Cisco Packet Tracer.

Theory

What is a hub?

A hub is a basic networking device used to connect multiple computers in a Local Area Network (LAN). It operates at the Physical Layer (Layer 1) of the OSI model. A hub does not filter or manage traffic; it simply acts as a central connection point. All devices connected to a hub share the same communication channel.

Explain the working principle of a hub.

A hub works by receiving data signals from one connected device and broadcasting them to all other devices connected to it. It does not examine MAC addresses or determine the destination of the data. Every device in the network receives the transmitted data, but only the intended recipient processes it. Since it broadcasts data blindly, network collisions are more likely in a hub-based network.

What is MAC address?

A MAC (Media Access Control) address is a unique hardware identification number assigned to a network interface card (NIC). It is used to identify devices within a local network. A MAC address operates at the Data Link Layer (Layer 2) of the OSI model. It is typically represented as a 12digit hexadecimal number (e.g., 00:1A:2B:3C:4D:5E). MAC addresses are permanently assigned by the manufacturer.

What is IP address?

An IP (Internet Protocol) address is a logical address assigned to a device for identification over a network. It allows devices to communicate across local and global networks such as the Internet. IP addresses operate at the Network Layer (Layer 3) of the OSI model. They can be either IPv4 (e.g., 192.168.1.1) or IPv6 format. Unlike MAC addresses, IP addresses can change depending on the network.

What is a Local Area Network (LAN)?

A Local Area Network (LAN) is a network that connects computers and devices within a limited geographical area such as a home, office, or school. It allows devices to share resources like files, printers, and internet connections. LANs typically use Ethernet cables or Wi-Fi for communication. They provide high-speed data transfer within the local environment.

What is a star topology?

Star topology is a network structure in which all devices are connected to a central device such as a hub or switch. Each device has a separate cable connection to the central point. If one cable fails, only that device is affected, not the entire network. However, if the central device fails, the whole network stops functioning. Star topology is widely used in modern LANs.

Hardware Components Required

- **PC-PT (x3):** End devices acting as network nodes.
- **Hub-PT:** A generic Physical Layer device used to interconnect the nodes.
- **Copper Straight-Through Cables:** The transmission media used for PC-to-Hub connectivity.

Hardware and Connection Setup

- **Network ID:** 192.168.1.0
- **Subnet Mask:** 255.255.255.0
- **PC Naming Convention:**
  - PC 1: **Batch_Roll_A** (e.g., S11_01_A)
  - PC 2: **Batch_Roll_B** (e.g., S11_01_B)
  - PC 3: **Batch_Roll_C** (e.g., S11_01_C)

Procedure

Step 1: Physical Topology Design

1. Open Cisco Packet Tracer and select **Network Devices > Hubs > Hub-PT**. Place it in the center.
2. Select **End Devices > PC** and place four units around the Hub.
3. **Naming:** Click on each PC label and rename them following the convention: S11_01_A, S11_01_B, S11_01_C and S11_01_D.

Step 2: Hardware Identification

1. Click on the **Hub** and select the **Physical** tab.
2. Observe the ports. Note that unlike the Switch, a Hub usually has fewer ports and no "intelligence" indicators (like console ports).

Step 3: Physical Cabling

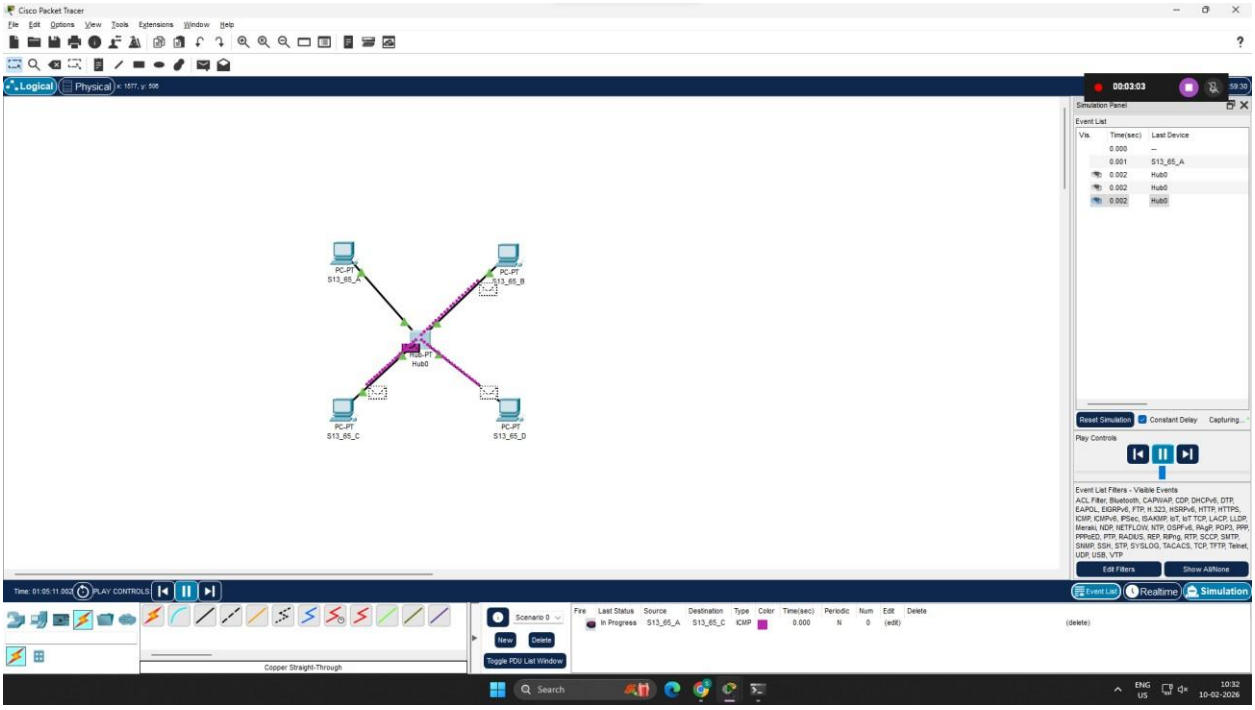
1. Select the **Copper Straight-Through** cable (solid black line).
2. Connect **S11_01_A (FastEthernet 0)** to **Hub (Port 0)**.
3. Repeat for the other three PCs using Hub ports **1, 2** and **3**.
4. **Observation:** Notice that the link lights turn **Green immediately**. Unlike a Switch, a Hub does not run STP (Spanning Tree Protocol), so there is no "Orange" waiting period.

Step 4: Logical Addressing (IP Configuration)

1. Click **S11_01_A > Desktop > IP Configuration**.
2. Enter Static IP: 192.168.1.1 | Subnet: 255.255.255.0.
3. Configure **S11_01_B** with IP 192.168.1.2, **S11_01_C** with IP 192.168.1.3 and **S11_01_D** with IP 192.168.1.4.

Step 5: Testing and Simulation (Crucial Step)

1. **Real-time:** Open the Command Prompt on **S11_01_A** and type ping 192.168.1.3.
2. **Simulation Mode:**
   - Click the **Simulation** tab and filter for **ICMP**. o Use the **Add Simple PDU** (Envelope) to click **S11_01_A** then **S11_01_C**.
   - Click **Play** and observe carefully: The packet goes to the Hub, and the Hub sends it to **all** PC B, PC C and PC D.
   - Observe the **Red Cross** on PC B and PC D (because they reject a packet not meant for them) and the **Green Check** on PC C.



6. Observations

- **Instant Connectivity:** Hub ports do not have a "learning" phase; they transition to green immediately upon connection.
- **Broadcast Behavior:** During simulation, the Hub was observed to forward the packet to every connected device. This wastes bandwidth and creates security risks.
- **Device Rejection:** End devices (PCs) are responsible for checking the destination MAC address. If it doesn't match, the PC discards the packet (indicated by the red cross in simulation).

Conclusion

The experiment successfully demonstrates the hardware and functional limitations of a Hub. It concludes that a **Hub** is a non-intelligent Layer 1 device that relies on broadcasting, making it less efficient than a Switch for modern Local Area Networks.